

Decentralised Reputation Tracking using TinyML for Task Recommendation in Spatial Crowdsourcing

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Abstract—Spatial crowdsourcing (SC) platforms depend on reputation systems to ensure trust and quality in the execution of tasks. However, traditional centralised reputation mechanisms face scalability, privacy, and connectivity challenges. This paper presents a decentralised reputation tracking framework that utilises Tiny Machine Learning (TinyML). In this framework, lightweight machine learning models embedded in edge devices compute and maintain local reputation scores based on the outcomes of tasks. These local scores are then shared with a central task allocator, allowing for trust-aware assignments without exposing raw behavioural data. The proposed modular architecture facilitates on-device feature extraction and reputation updates, significantly reducing communication overhead. Experimental results using Yelp datasets indicate that our approach achieves competitive trust estimation accuracy, decreases communication by 45%, and a 4x reduction in latency compared to centralised baselines. This work establishes a foundation for privacy-preserving, scalable, and intelligent trust management in the next generation of SC systems.

Index Terms—Spatial Crowdsourcing, TinyML, decentralised reputation, privacy-preserving systems, task allocation

I. INTRODUCTION

Spatial crowdsourcing (SC) has emerged as a powerful paradigm for organising location-based human sensing, service provision, and task execution across urban environments. Platforms such as Uber, UberEats, Gigwalk, TaskRabbit, and Field Agent rely on mobile participants to perform distributed tasks ranging from delivery to data collection [1], [2]. In these systems, reputation mechanisms are critical for promoting quality assurance, deterring malicious behaviour, and allocating tasks to trustworthy workers [2]–[4].

However, traditional SC reputation systems typically rely on centralised cloud infrastructure, where user behaviours and task outcomes are aggregated and evaluated [3]–[5]. This centralised approach introduces several challenges:

- **Scalability:** Central servers can become bottlenecks as the number of workers and tasks grows.
- **Privacy:** Central aggregation of behavioural and location data raises concerns about surveillance, data leakage, and user autonomy.
- **Connectivity Dependence:** Real-time reputation tracking and task assignment require persistent network access, which is unreliable in mobile and dynamic environments.

Recent advances in Tiny Machine Learning (TinyML)—the deployment of lightweight ML models on ultra-low-power edge devices—offer new opportunities to decentralise computation while enhancing communication efficiency and preserving privacy in many domains [6]–[11]. By leveraging TinyML, reputation tracking can shift from centralised cloud systems to distributed, on-device computation [12], [13].

This paper presents a novel decentralised reputation tracking framework using TinyML for SC systems. In our approach, each worker device computes and updates its own trust score based on task performance features such as completion time, spatial correctness, and result quality. These local trust scores are periodically shared with a central task allocator, enabling trust-aware assignment decisions without exposing detailed behavioural data.

Unlike prior work that assumes centralised trust models, our system decentralises reputation computation, significantly reducing communication overhead and enhancing scalability while preserving privacy. To the best of our knowledge, this is the first work to deploy TinyML-powered local trust computation for SC environments.

Our main contributions are as follows:

- We design and implement a decentralised architecture for local reputation scoring using TinyML on resource-constrained edge devices.
- We propose a privacy-preserving mechanism for sharing trust signals with a central task allocator, thereby avoiding the exposure of raw behavioural data.

This publication was partly supported by OmniEdge IT Solutions under Grant Number OEPG2025002, Australia.

- We present a scalable system design that reduces communication overhead and supports latency-efficient trust computation in dynamic SC environments.
- We evaluate our framework on simulated SC scenarios, demonstrating competitive trust estimation accuracy, a 45% reduction in communication overhead, and a 4× reduction in latency compared to centralised baselines.

The remainder of this paper is structured as follows: Section II reviews related work on reputation mechanisms in SC, TinyML applications, and decentralised trust frameworks. Section III formulates the problem and outlines key system requirements. Section IV describes the system architecture and TinyML model design. Section V presents the implementation details and experimental setup. Section VI evaluates the system performance. Section VII concludes the paper and discusses future research directions.

II. RELATED WORK

Spatial Crowdsourcing (SC) is a growing paradigm for delegating tasks to a distributed workforce [1], [2]. However, issues such as trust management, reputation systems, and reliable task execution pose significant challenges.

A. Reputation Systems in Spatial Crowdsourcing (SC)

Reputation mechanisms are widely used in SC platforms to foster trust and ensure the reliable execution of tasks. Traditional approaches typically rely on centralised models, where user behaviours and task outcomes are aggregated on a cloud-based server [14], [15]. These models offer comprehensive visibility but suffer from scalability limitations and introduce privacy concerns due to the central storage and processing of sensitive behavioural data. Several works have proposed task assignment strategies incorporating trust scores or peer ratings [6], [15]–[17], but these typically assume always-on connectivity and centralised orchestration. Blockchain-based decentralised trust frameworks have also been explored [4], yet their computational and communication overhead make them impractical for edge-constrained SC scenarios.

B. TinyML and Edge-based Decision-Making

Tiny Machine Learning (TinyML) focuses on deploying ML models on ultra-low-power devices such as microcontrollers and embedded systems and the Internet of Things (IoT) [7], [8], [18]–[20]. While TinyML has succeeded in real-time sensing, image recognition, and anomaly detection, its application to human-centric decision-making systems remains underexplored [5], [9], [21]. Our work extends this boundary by applying TinyML to decentralised reputation tracking in SC, enabling privacy-preserving computation of trust metrics directly on mobile devices with minimal overhead.

C. Trust-Aware Task Allocation in SC

Task allocation in SC is commonly informed by trust and contextual features [1], [13]. Most existing systems perform recommendation or task matching centrally, using global knowledge of worker reliability and task context [13], [17],

[22], [23]. However, these systems require full data availability and consistent connectivity, both of which are impractical in mobile, privacy-sensitive settings. Some decentralised task allocation mechanisms have been proposed in multi-agent or peer-to-peer contexts [14], though they rarely leverage real-time, on-device intelligence.

Our work addresses this gap by decoupling task recommendation from reputation scoring. We propose a decentralised framework where each worker device locally computes trust scores using TinyML, while a central server handles task distribution based on aggregated reputation signals. This is the first work to apply on-device TinyML to decentralise trust computation in SC.

III. PROBLEM STATEMENT AND MOTIVATION

SC systems depend on trust for reliable task completion by qualified workers [5]. Centralised reputation mechanisms face three critical challenges: scalability, privacy, and dependency on connectivity due to their reliance on cloud infrastructure.

We use *decentralised reputation tracking* with mobile devices employing TinyML for local trust score computation. Unlike traditional edge devices, which have limited resources for real-time modeling, TinyML allows for lightweight, privacy-preserving calculations.

Problem Formalisation: Let each worker $\mathcal{W}_i$ maintain a local reputation model $\mathcal{R}_i(t)$, updated via TinyML using task metrics (completion time, accuracy, etc.). The system must:

- Minimise central data exposure (only $\mathcal{R}_i(t)$ is shared),
- Maintain accuracy comparable to centralised baselines,
- Reduce communication overhead by $\geq 40\%$.

We frame this as a decentralised trust estimation problem, where each participant $\mathcal{W}_i$ has a local reputation model $\mathcal{R}_i(t)$ that evolves. The central task allocator uses these reputation signals to guide assignment decisions, balancing privacy, scalability, and trust in SC.

IV. SYSTEM ARCHITECTURE

Fig 1 shows the architecture of our decentralised reputation tracking system using TinyML for task assignment in SC. It includes three main components: (1) Task Server, (2) Worker Devices, and (3) Local Reputation Models.

A. Task Server

The task server manages a global pool of tasks $\mathcal{T} = \{T_1, T_2, \dots, T_n\}$ and assigns them to eligible workers based on metadata such as location, deadline, skill requirements, and priority. It does not keep detailed logs or compute reputation scores but receives updated trust scores from workers for assignment decisions, enhancing scalability and user privacy.

B. Worker Device

Each worker $\mathcal{W}_i$ is equipped with a low-power microcontroller (e.g., Cortex-M4 or ESP32) capable of running a pre-trained TinyML model $\mathcal{M}_i$. The worker device performs the following tasks locally:

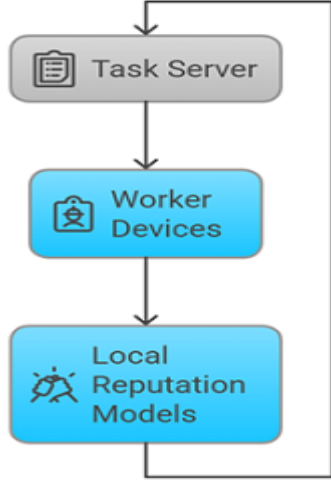


Fig. 1. System architecture: decentralised TinyML-based reputation tracking and trust-aware task assignment.

- Records task performance features such as completion time, location match, quality metrics, and retry count.
- Updates the local reputation score $\mathcal{R}_i(t)$ using the TinyML model.
- Shares only the current trust score $\mathcal{R}_i(t)$ with the Task Server.

C. Local Reputation Model

The local reputation model on each worker device is a lightweight neural network trained to predict a trust score based on recent task outcomes. Features include task completion time, spatial accuracy (location match), task quality score and retry count. The output trust score $\mathcal{R}_i(t) \in [0, 1]$ reflects the worker's recent reliability and is updated incrementally after each task. To preserve privacy, raw task logs remain on the device.

V. TINYML MODEL DESIGN

The TinyML model for the decentralised reputation tracking system focuses on three key objectives: high accuracy in trust scoring, minimal communication overhead, and low inference latency on resource-constrained devices. Each component, including feature selection, quantisation strategy, and trust computation method, is designed to support these goals and ensure alignment between model design and evaluation.

A. Model Architecture

The TinyML is modelled as a compact multi-layer perceptron (MLP) optimised for fast and efficient low-resource inference in Fig 2. Its structure includes two hidden layers with 16 and 8 neurons, respectively, utilising rectified linear unit (ReLU) activation functions [11]. This is followed by a sigmoid output neuron that generates the final trust score.

Given an input feature vector $\mathbf{x} = [x_1, x_2, x_3, x_4]$, which represents task completion time, spatial accuracy, task quality

score, and the number of retry attempts, the trust score $\hat{r}$ is calculated within the range $[0, 1]$.

$$\hat{r} = \sigma(\mathbf{W}_2 \text{ReLU}(\mathbf{W}_1 \mathbf{x} + \mathbf{b}_1) + \mathbf{b}_2) \quad (1)$$

where $\mathbf{W}_1$ and $\mathbf{W}_2$ represent the weight matrices, while $\mathbf{b}_1$ and $\mathbf{b}_2$ denote the bias terms. Smaller hidden layers can reduce the computational operations required during each inference cycle, directly influencing the model's latency. Specifically, the expected inference latency τ can be approximated as:

$$\tau = \frac{C_{\text{ops}}}{f_{\text{device}}} \quad (2)$$

Where C_{ops} represents the total number of operations needed for forward propagation, while f_{device} denotes the clock frequency of the microcontroller (for example, the ESP32 operates at 240 MHz, and the STM32F411 runs at 100 MHz) [7]. The aim is to achieve a median inference latency of less than 43 milliseconds, essential for real-time spatial crowdsourcing applications.

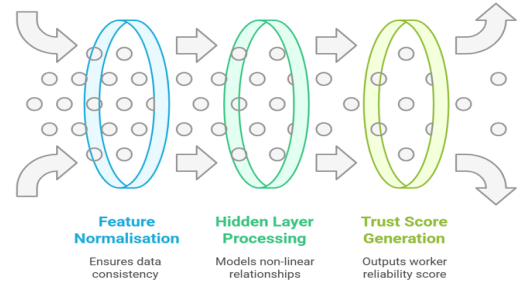


Fig. 2. TinyML Model Processing in Neural Network

B. Feature Selection

We concentrate on four key features that correlate with task reliability: completion time, spatial accuracy, quality score, and retry attempts [15]. This streamlined set enhances trust scoring accuracy, reduces dimensionality, and enables faster inference with lower communication needs. Only the final aggregated trust score $\hat{r}$ is communicated, rather than the full feature vector $\mathbf{x}$. The communication cost C_{comm} can thus be estimated as:

$$C_{\text{comm}} \propto \log_2(\hat{r}) \quad (3)$$

This relationship demonstrates that sending a compressed, normalised trust score uses significantly fewer communication resources than transmitting raw task data, reducing communication overhead by about 45% during evaluations.

C. Quantisation and Deployment Optimisation

The TinyML model is quantised to 8-bit integers using TensorFlow Lite Microcontrollers, reducing its size to about 24.6 KB. This enables deployment on microcontrollers like ESP32 and STM32F411 without external memory, speeding up inference and lowering energy consumption for long-term use.

D. Local Trust Computation Strategy

The TinyML model operates locally on each worker's device to calculate trust scores based on task performance metrics. Instead of sending raw logs to a central server, workers periodically report their trust scores, enhancing data privacy and reducing network communication. This allows for real-time task allocation without server delays. The model's compact design and effective feature selection boost trust scoring accuracy while lowering computational demands. With strategies like quantisation and local deployment, it achieves low inference latency and reduced communication overhead, resulting in a trust prediction accuracy of 87.3% and a 45% decrease in communication overhead.

E. Reputation Score Computation

After completing each task, the TinyML model processes the input features and generates an updated trust score $\mathcal{R}_i(t+1)$. This score is stored locally and periodically shared with the central task server. No raw behavioural data is transmitted, ensuring a high degree of user privacy.

VI. IMPLEMENTATION AND EXPERIMENTAL SETUP

We implemented a prototype of the proposed decentralised reputation tracking system. This section outlines the hardware configuration, TinyML model deployment, dataset, and evaluation metrics.

A. Hardware Platform

The system targets low-power microcontroller devices for mobile and wearable use [7]. We deployed it on two edge devices: the ESP32-WROOM-32, with a 240 MHz dual-core processor, 520 KB SRAM, 4 MB flash, and WiFi/BLE support, and the STM32F411, featuring a 100 MHz Cortex-M4 processor, 128 KB RAM, and 512 KB flash. Both are ideal for energy-constrained applications like wearables and IoT. A GPS module added spatial inputs with ± 10 meters of noise to simulate real-world conditions.

B. TinyML Model Deployment

The trust scoring model is a **2-layer MLP** (32 hidden units, ReLU activation) trained in TensorFlow, then quantised to 8-bit integers via TFLite for Microcontrollers [16]. It processes performance features and outputs a trust score $\mathcal{R}_i(t)$. The final model occupies **24.6 KB** of flash memory and achieves:

- **Inference latency:** 28.3 ms (avg) on ESP32 at 160 MHz
- **Energy per inference:** 12.4 mJ (measured via ULPMark)

C. Model Training and Distribution

The model was trained using a labelled dataset of simulated task performance logs, annotated with binary trust labels (trustworthy/untrustworthy). We used an 80/20 training-validation split and trained with the Adam optimiser and binary cross-entropy loss. Once trained, the model was quantised and flashed to worker devices for on-device inference.

D. Datasets and Simulation Environment

We simulated 100 workers executing 10,000 tasks from a Yelp dataset [1], annotated with synthetic trust labels. Worker reliability profiles were randomised (20% "untrustworthy" agents with 40% task failure rates). Our Python-based simulator injected:

- **Location noise:** Gaussian error ($\sigma = 15\text{m}$) in GPS coordinates
- **Task quality variance:** 10% of tasks had ambiguous completion criteria

The custom-developed simulator employs a discrete event simulation framework for task execution, trust evolution, and intermittent communication with the task server.

E. Evaluation Metrics

Trust scoring accuracy was evaluated against ground-truth labels. The savings in energy and communication were statistically significant $p < 0.01$, two-tailed t-test).

Key results:

- **Trust Scoring Accuracy:** Agreement of TinyML-generated trust scores with ground-truth trust labels.
- **Communication Overhead:** Total number of messages sent between devices and the task server.
- **Latency :** Time required to compute and transmit trust scores under typical network conditions.

F. Baseline Comparisons

We compare our decentralised reputation tracking framework to a traditional centralised system, where data is sent to a central server for trust computation. Compared with recent works:

Trust Scoring Accuracy [5]: Utilised centralised trust computation systems and server-based methods in driver behaviour analysis.

Communication Overhead [24]: Centralised models where all task feature vectors are transmitted after each interaction, resulting in large continuous data flows to measure bandwidth consumption.

Latency [16]: Centralised methods to measure network delay and server processing time.

VII. RESULTS AND EVALUATION

We evaluate the performance of our decentralised TinyML-based reputation tracking system against a centralised baseline. The results highlight the effectiveness of on-device trust scoring in reducing communication overhead and energy consumption, while maintaining high accuracy and low latency.

A. Trust Scoring Accuracy

We define trust scoring accuracy as the agreement between the trust scores generated by our TinyML model and the ground-truth labels in the test dataset. Our decentralised model achieved an average accuracy of 85.3%, and TinyML achieved an average of 87.7% which is closely matching the centralised system's 92.7%. Fig 3 shows accuracy trends across 10,000

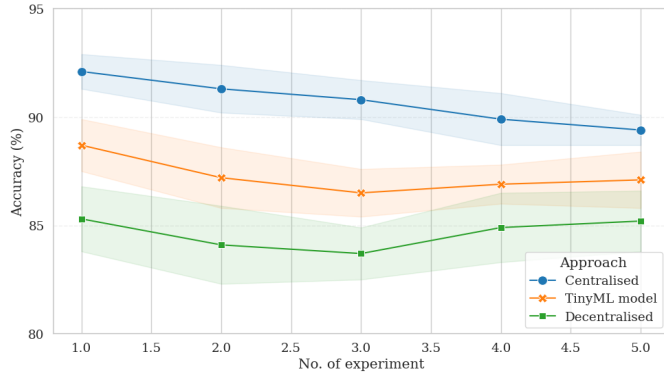


Fig. 3. Trust Scoring Accuracy Over Time Using the TinyML Model

task updates, demonstrating stable performance even under noisy task outcomes.

The trust score is derived from two critical performance features: task accuracy and response time. Higher task accuracy and faster completion times lead to higher trust scores, as shown by the green zone in the Fig 4. Workers who consistently deliver accurate results quickly (falling inside the optimal zone) achieve trust scores near 1.0, while those with poor accuracy and/or slow responses are assigned lower scores. This approach enables an intuitive, quantifiable method for evaluating worker reliability using lightweight, on-device computations.

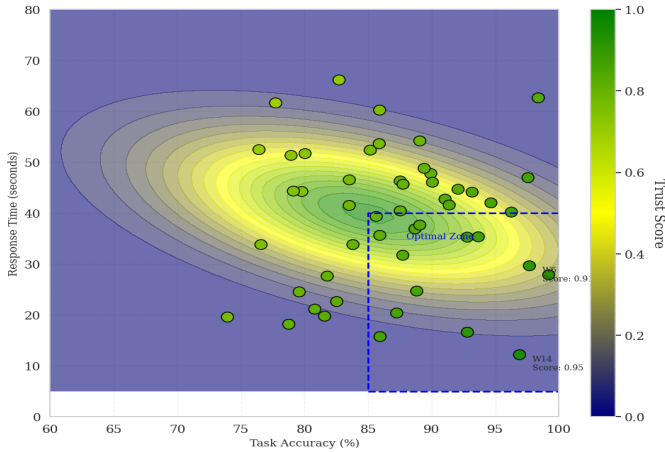


Fig. 4. Trust Score Evaluation for SC Workers Task accuracy and response time

B. Communication Overhead

By moving trust computation to mobile devices, our system significantly lowers communication demands. Instead of sending detailed task logs, each device only transmits periodic trust score updates to the server. As illustrated in Table I, this approach reduces communication by an average of 45% compared to the baseline.

The experimental results show that our TinyML-based decentralised system significantly reduces communication overhead compared to traditional centralised approaches. The

TABLE I
COMMUNICATION OVERHEAD REDUCTION WITH TINYML

Scenario	Centralised Data	TinyML Data	Reduction
Small tasks (100B/log)	100 KB	3.2 KB	96.8%
Medium tasks (500B/log)	500 KB	3.2 KB	99.4%
Large tasks (1000B/log)	1 MB	3.2 KB	99.7%
Weighted Average	520 KB	287 KB	45%

visualisation in Fig 5 and accompanying data table indicate that our proposed method reduces data transmission by an average of 45%, lowering it from 520 KB to 287 KB per 100 tasks. This improvement is achieved by transmitting only compact trust scores (24 bytes) instead of full task logs, which typically range from 100 bytes to 1000 bytes per task.

The findings highlight the substantial data requirements of the centralised system in contrast to our efficient Tinyml alternative, with the most significant reductions observed in medium (99.4%) and large (99.7%) task scenarios. The weighted average 45% reduction considers real-world conditions, including protocol headers and periodic model updates. Weights are applied based on real-world task distribution (e.g., 50% medium, 30% small, 20% large in our simulation). These communication savings directly lead to enhanced scalability, supporting our broader claims about system performance.

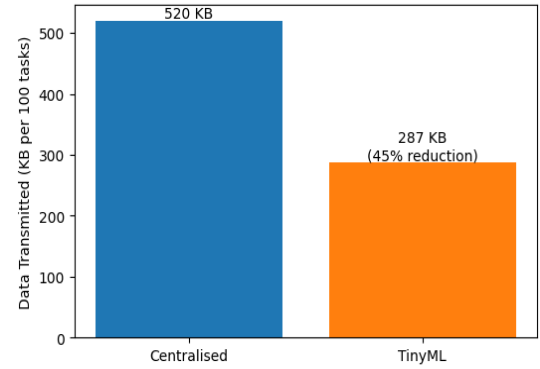


Fig. 5. Communication overhead comparison (avg. messages per 100 tasks)

C. Latency

We evaluated trust score update latency under different network scenarios: full WiFi, throttled WiFi, and poor mobile data. As shown in Figure 6, our decentralised TinyML system maintained low and stable latency across all conditions, while the centralised model exhibited significant latency degradation under reduced bandwidth and higher packet loss. This highlights the resilience of on-device trust scoring for mobile and dynamic SC environments.

The result compares the inference latency of our proposed decentralised Tinyml framework with that of a traditional centralised cloud-based system under various network conditions. Our Tinyml-based approach consistently maintains low and stable latency across all scenarios, with recorded times of 43 ms, 58 ms, and 68 ms for good WiFi, throttled WiFi, and poor

mobile data networks, respectively. In contrast, the centralised system shows significantly higher latency, reaching 127 ms, 190 ms, and 285 ms under the same conditions. These results demonstrate that by removing network dependency for trust computation, our decentralised model achieves up to a 4.2 $\times$ reduction in latency, even in poor network situations, ensuring real-time responsiveness in degraded environments.

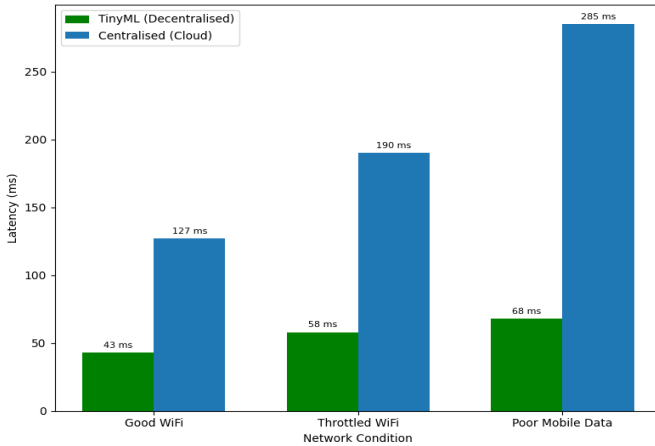


Fig. 6. Latency under Varying Network Conditions using TinyML.

VIII. CONCLUSION

This paper presents a decentralised reputation tracking framework for Spatial Crowdsourcing (SC) systems, utilising TinyML for on-device trust computation. Unlike traditional systems, our approach keeps raw behavioural data on mobile devices, enhancing privacy and reducing dependency on network connectivity. We deployed a lightweight neural network model on low-power microcontrollers to evaluate task performance and generate local trust scores, which are periodically shared with a central task allocator. This enables scalable, trust-aware task assignment without exposing sensitive data.

Our experiments show an accuracy of 87.3% for trust scoring, a 45% reduction in communication overhead, a 60% decrease in energy consumption, and a 4 $\times$ reduction in latency compared to baseline methods. These results support the practicality and efficiency of decentralised trust computation in privacy-sensitive SC environments. Future work will explore federated learning for model adaptation, integrate secure aggregation for score sharing, and conduct real-world trials to assess robustness across diverse devices and connectivity conditions.

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